

This sounds like a great assignment to dive into Natural Language Processing (NLP). Since you're exploring how search engines and recommendation systems actually "understand" what a document is about, **TF-IDF (Term Frequency-Inverse Document Frequency)** is the perfect place to start.

Here is a breakdown of how the logic works and a walkthrough of how you can identify those top keywords.

1. Understanding the Mechanics

TF-IDF isn't just a fancy acronym; it's a statistical measure used to evaluate how important a word is to a document in a collection. It consists of two parts:

- **Term Frequency (TF):** How often a word appears in a single document. If "Algorithm" appears 10 times in Document A, it's likely important.
- **Inverse Document Frequency (IDF):** How unique the word is across all 5 documents. If the word "The" appears in every document, its importance score drops to near zero.

Mathematically, it looks like this:

$$W_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

Where:

- $W_{i,j}$ is the weight of term i in document j .
- N is the total number of documents (5 in your case).
- df_i is the number of documents containing term i .

2. Steps to Identify Top Keywords

To complete your assignment, follow this logical flow:

1. **Preprocessing:** Clean your 5 documents. Convert text to lowercase, remove punctuation, and strip out "stop words" (like *and*, *but*, *the*) which carry no real meaning.
 2. **Vectorization:** Apply the TF-IDF algorithm. You can do this manually for a small set, but using a library like Python's `scikit-learn` makes it much faster.
 3. **Ranking:** For each document, sort the words by their TF-IDF score. The highest scores are your **keywords**.
 4. **Extraction:** Select the top 3–5 words per document.
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3. How to Explain the Results

When you identify your keywords, your explanation should justify *why* the math chose them. Here is a template you can use for your 5 documents:

Document	Top Keyword	Explanation
Doc 1	<i>Photosynthesis</i>	High TF within this doc; low IDF because other docs discuss different topics.
Doc 2	<i>Blockchain</i>	Frequent in this specific text; unique compared to the rest of the corpus.
Doc 3	<i>Renaissance</i>	Identified as a "signature" word that distinguishes this doc from the others.

4. Why this matters

The beauty of TF-IDF is that it filters out common language and highlights the **distinctive** vocabulary. If you see a word like "Data" as a top keyword, it means that specific document is heavily focused on data while the others are not.

Would you like me to generate some sample Python code using `TfidfVectorizer` so you can run this on your 5 documents?